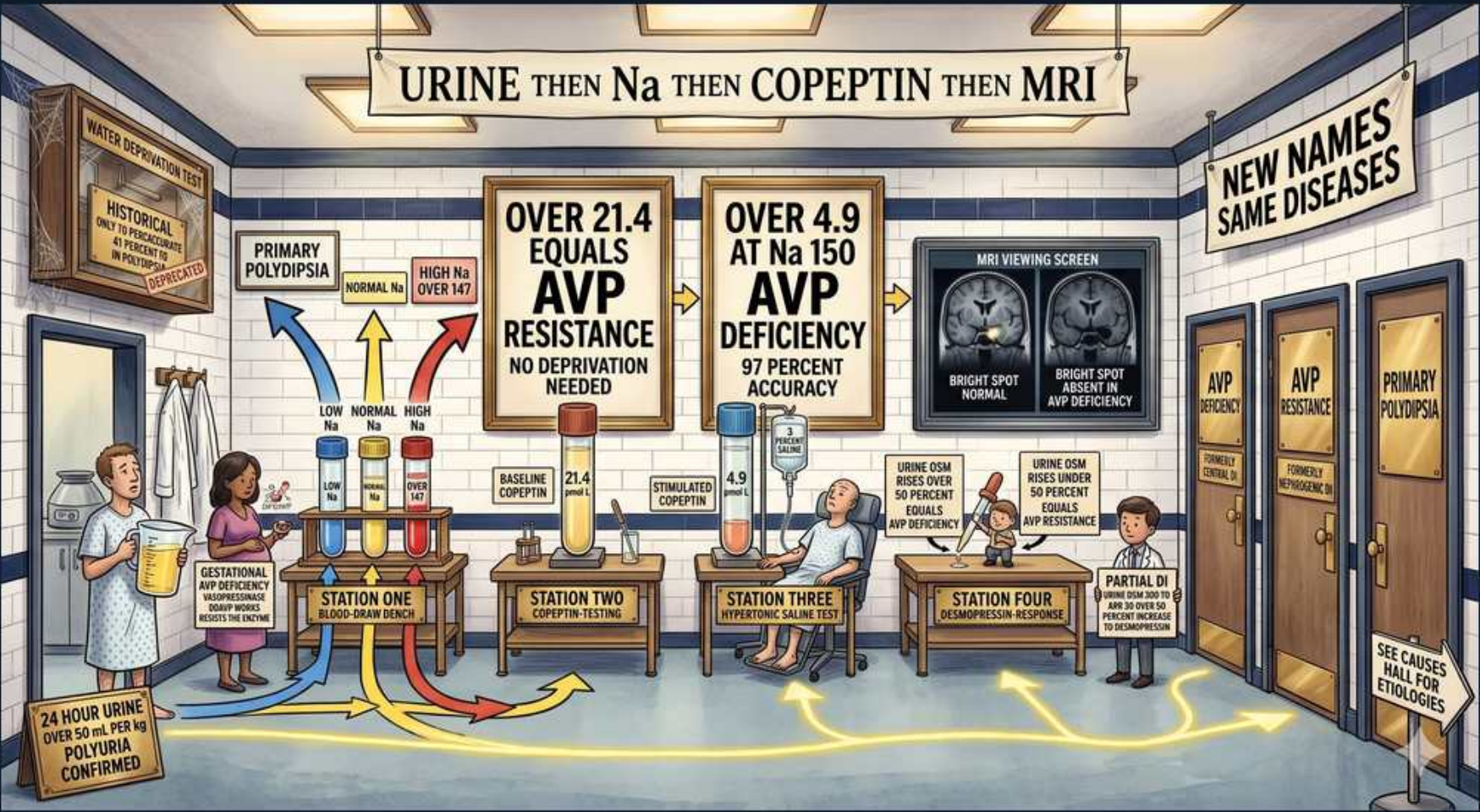


Pituitary/DI/SIADH — Polyuria diagnostic approach



1. Urine then Na then copeptin then MRI

WHAT YOU SEE

A wooden hanging banner across the top reads URINE then Na then COPEPTIN then MRI — the banner literally spells the four-step order of the polyuria work-up left to right.

WHAT IT ENCODES

Stepwise polyuria workup: (1) confirm true polyuria by 24-h urine volume, (2) measure serum Na/osmolality and urine osmolality to separate water from osmotic diuresis, (3) measure copeptin — baseline and hypertonic-saline-stimulated — to type the DI, and (4) MRI the pituitary for structural disease and the bright spot. Modern practice uses copeptin in place of the cumbersome classic water-deprivation test.

BOARD TRAP

WRONG: start with an empiric desmopressin trial before any labs. Discriminator — giving DDAVP first can dangerously lower sodium in a primary-polydipsia patient and obscures the diagnosis; you must first confirm polyuria and characterize the diuresis biochemically.

2. 24h urine >3L confirms polyuria

WHAT YOU SEE

Bottom-left: 24-hour urine over 50 ml/kg polyuria confirmed

WHAT IT ENCODES

Polyuria is defined as urine output >3 L/day (or >50 mL/kg/day) — this distinguishes true polyuria from urinary frequency or nocturia with normal total volume. Confirming the volume is the essential first step before any DI testing, since 'frequency' from a small overactive bladder is a different problem entirely.

BOARD TRAP

WRONG: urinary frequency without high total volume defines polyuria. Discriminator — polyuria requires a high 24-h VOLUME (>3 L); frequent small voids (overactive bladder, UTI, BPH) are frequency, not polyuria, and do not need a DI workup.

3. Osmotic vs water diuresis

WHAT YOU SEE

Top-left wall plaque labeled WATER DEPRIVATION TEST with a big DEPRECATED stamp across it — the crossed-out classic test is the springboard for splitting polyuria into osmotic vs water diuresis before reaching for copeptin.

WHAT IT ENCODES

Split polyuria by urine osmolality: OSMOTIC diuresis has a high urine solute output (Uosm typically >300, e.g., uncontrolled diabetes/glucosuria, mannitol, high-protein tube feeds, post-obstructive urea) while WATER diuresis has dilute urine (Uosm <300). Both DI subtypes and primary polydipsia are water diuresis; identifying osmotic diuresis (check glucose first) avoids an unnecessary DI workup.

BOARD TRAP

WRONG: all polyuria represents a water diuresis. Discriminator — a large share is OSMOTIC (most commonly hyperglycemic glucosuria); a high Uosm with high solute excretion points to osmotic diuresis, not DI, so always check glucose/solute load before chasing AVP.

4. Primary polydipsia / low Na

WHAT YOU SEE

Sign: primary polydipsia, low Na <147 / high Na >147

WHAT IT ENCODES

Baseline serum sodium triages the water-diuresis causes: LOW-normal Na (overdrinking dilutes plasma) suggests primary polydipsia, whereas HIGH or high-normal Na (renal water loss outpacing intake) suggests true DI. A frankly elevated baseline Na essentially excludes primary polydipsia.

BOARD TRAP

WRONG: a high serum sodium points to primary polydipsia. Discriminator — primary polydipsia DILUTES the plasma (low-normal Na); high Na reflects water LOSS and indicates true DI (central or nephrogenic). This single value is one of the most useful early discriminators.

5. Station 1: copeptin testing

WHAT YOU SEE

The first lab bench, STATION ONE BLOOD-DRAW BENCH, with colored blood tubes — this is where the copeptin sample is drawn, the entry point of the algorithm.

WHAT IT ENCODES

Copeptin — the stable surrogate co-secreted 1:1 with AVP — has largely replaced the indirect water-deprivation test, which was slow and frequently misclassified partial cases. Copeptin can be assayed at baseline and after osmotic stimulation, giving a more direct readout of AVP secretory capacity.

BOARD TRAP

WRONG: copeptin cannot distinguish the DI subtypes. Discriminator — copeptin DOES separate them: high baseline indicates nephrogenic DI (resistance with high endogenous AVP), and stimulated copeptin separates central DI from primary polydipsia with high accuracy.

6. Baseline copeptin >21.4

WHAT YOU SEE

Sign: over 21.4 equals AVP resistance, no deprivation needed

WHAT IT ENCODES

A baseline (unstimulated) copeptin >21.4 pmol/L diagnoses NEPHROGENIC DI (AVP resistance) outright — the kidney isn't responding, so endogenous AVP/copeptin is already very high — and NO water deprivation or saline stimulation is needed. This is the quick exit from the algorithm at the very first measurement.

BOARD TRAP

WRONG: a high baseline copeptin means central DI. Discriminator — central DI is AVP DEFICIENCY, so copeptin is LOW; a markedly high baseline copeptin reflects resistance = NEPHROGENIC DI. Don't invert the deficiency vs resistance logic.

7. Station 2: baseline copeptin

WHAT YOU SEE

STATION TWO COPEPTIN TESTING bench holding a tube marked 21.4 — the baseline copeptin draw, where a value over the 21.4 cutoff stops the work-up at nephrogenic DI.

WHAT IT ENCODES

The sequence is to draw baseline copeptin FIRST: if it is >21.4 pmol/L you have nephrogenic DI and can stop; if it is below that cutoff (non-diagnostic), proceed to a hypertonic-saline-stimulated copeptin to separate central DI from primary polydipsia. This tiered approach minimizes the need for the more demanding stimulation test.

BOARD TRAP

WRONG: you must perform hypertonic-saline stimulation in every patient before any diagnosis. Discriminator — a high BASELINE copeptin (>21.4) already diagnoses nephrogenic DI, sparing the patient stimulation; stimulation is reserved for indeterminate baselines.

8. Stimulated copeptin >4.9

WHAT YOU SEE

Sign: over 4.9 at Na 150, 97 percent accuracy

WHAT IT ENCODES

After hypertonic-saline infusion raises serum Na to ~150 mmol/L, a stimulated copeptin >4.9 pmol/L indicates PRIMARY POLYDIPSIA (the AVP axis responds appropriately to the osmotic stress), whereas ≤4.9 indicates CENTRAL DI (the axis fails to respond). This cutoff distinguishes the two with ~97% accuracy, far better than the old water-deprivation test.

BOARD TRAP

WRONG: a low stimulated copeptin means nephrogenic DI. Discriminator — nephrogenic DI is already settled by a high BASELINE copeptin; the STIMULATED test separates central DI (low response, ≤4.9) from primary polydipsia (good response, >4.9).

9. Station 3: hypertonic saline

WHAT YOU SEE

STATION THREE HYPERTONIC SALINE TEST bench with a hanging IV saline bag feeding a seated patient and a tube marked 4.9 — the 3% saline infusion that drives Na to 150 and stimulates copeptin.

WHAT IT ENCODES

The hypertonic-saline test infuses 3% saline to drive serum Na up to ~150 mmol/L, maximally stimulating AVP/copeptin to reveal secretory capacity. It is the discriminating step between central DI and primary polydipsia and requires close, frequent sodium monitoring during the infusion to do it safely.

BOARD TRAP

WRONG: the hypertonic-saline copeptin test can be run without monitoring sodium during the infusion. Discriminator — you are intentionally raising Na to ~150, so frequent Na checks are mandatory to avoid overshoot/hypermnatremic injury; it is a monitored procedure, not a simple blood draw.

10. Station 4: desmopressin response

WHAT YOU SEE

STATION FOUR DESMOPRESSIN-RESPONSE bench with a dropper held by a small figure — the DDAVP challenge station where urine osmolality is rechecked after giving the hormone.

WHAT IT ENCODES

The desmopressin challenge is the functional confirmatory step: give DDAVP and recheck urine osmolality. Central DI urine concentrates briskly (Uosm rises >50%) because the kidney can respond once hormone is supplied; nephrogenic DI shows little/no rise (resistance). Primary polydipsia shows only a modest rise (medullary gradient washed out).

BOARD TRAP

WRONG: nephrogenic DI concentrates the urine well after desmopressin. Discriminator — nephrogenic DI is renal RESISTANCE, so DDAVP produces little/no rise in Uosm; a robust >50% rise after DDAVP indicates CENTRAL DI (deficiency that responds to replacement).

11. Uosm rises >50% = AVP deficiency

WHAT YOU SEE

Sign: urine Uosm rises over 50 percent equals AVP deficiency

WHAT IT ENCODES

A urine-osmolality rise of >50% after desmopressin confirms CENTRAL DI (AVP deficiency): the concentrating machinery is intact and just needed the hormone. This functional response, together with low baseline/stimulated copeptin, nails the central diagnosis and predicts a good response to DDAVP therapy.

BOARD TRAP

WRONG: a >50% rise in Uosm after DDAVP indicates nephrogenic DI. Discriminator — a strong response means the kidney CAN respond once given hormone = central (deficiency) DI; nephrogenic DI (resistance) by definition fails to respond.

12. Uosm <50% rise = AVP resistance

WHAT YOU SEE

Sign: urine Uosm rises under 50 percent equals AVP resistance

WHAT IT ENCODES

A urine-osmolality rise of <50% (or essentially none) after desmopressin confirms NEPHROGENIC DI (AVP resistance) — the collecting duct cannot respond despite adequate hormone. This matches a high baseline copeptin and steers treatment toward thiazide + low-salt diet ± amiloride/NSAID and removing the cause, NOT DDAVP.

BOARD TRAP

WRONG: a minimal Uosm rise after DDAVP points to central DI. Discriminator — failure to concentrate despite exogenous DDAVP defines RESISTANCE = nephrogenic DI; central DI is precisely the form that DOES concentrate after DDAVP.

13. MRI bright spot

WHAT YOU SEE

MRI viewing screen: bright spot normal vs absent

WHAT IT ENCODES

Pituitary MRI serves two roles: the normal posterior-pituitary T1 'bright spot' (stored AVP) is ABSENT in central DI but typically present in nephrogenic DI/primary polydipsia, and MRI screens for structural causes (tumor, stalk thickening from infiltrative disease, infundibuloneurohypophysitis). Absence of the bright spot supports central DI but is not by itself diagnostic (can be absent in some normal elderly).

BOARD TRAP

WRONG: an absent bright spot means nephrogenic DI. Discriminator — loss of the bright spot reflects depleted AVP STORES = CENTRAL DI; nephrogenic DI has intact synthesis/storage and usually a preserved bright spot.

14. New names same diseases

WHAT YOU SEE

Right doors: AVP resistance / AVP deficiency / primary polydipsia

WHAT IT ENCODES

Updated nomenclature (to prevent confusion with diabetes mellitus): central DI = 'AVP deficiency,' nephrogenic DI = 'AVP resistance,' and primary polydipsia keeps its name. The underlying diseases and management are unchanged — only the labels emphasize the deficiency-vs-resistance mechanism.

BOARD TRAP

WRONG: 'AVP resistance' is the new name for central DI. Discriminator — AVP RESISTANCE = nephrogenic DI (kidney can't respond), whereas AVP DEFICIENCY = central DI (not enough hormone). Mixing the two reverses both diagnosis and treatment.